Altera Cyclone IV EP4CE6 Development Board (V3.0)

Hardware Architecture & Peripheral Pin Reference

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1 System Architecture Overview

The Cyclone IV V3.0 board is built around an Altera EP4CE6 FPGA in a 144-pin Quad Flat Package (QFP). To support standalone operation without requiring external peripheral shields, the board integrates essential human-machine interface (HMI) components, secondary non-volatile flash storage, and regulated low-dropout (LDO) power supplies.

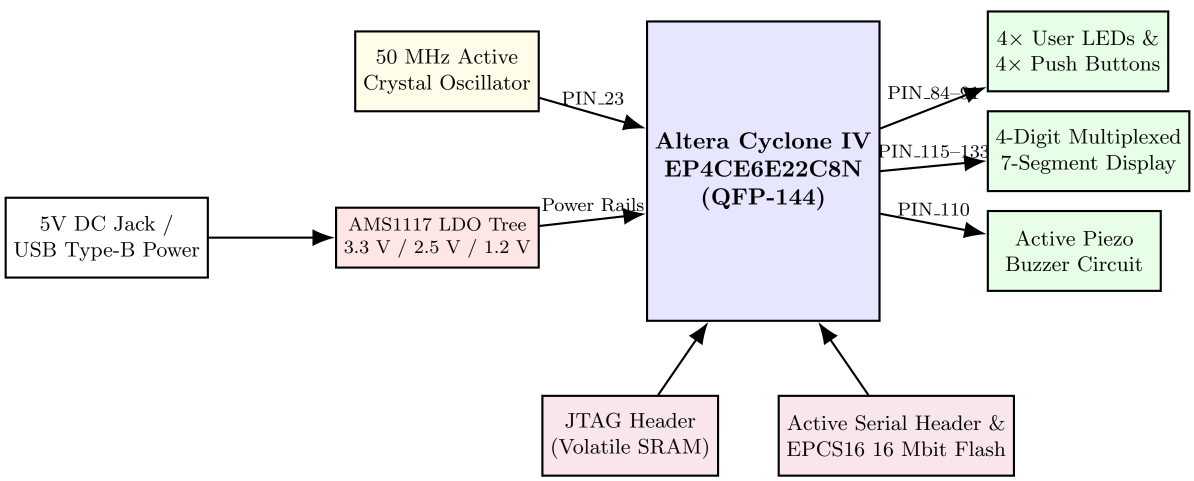

Figure 1: System-level hardware block diagram of the Altera Cyclone IV V3.0 development board.

2 Core FPGA Specifications (EP4CE6E22C8N)

The target FPGA belongs to the low-power Cyclone IV E series. The device nomenclature decodes as follows: **EP4C** (Cyclone IV architecture), **E** (Enhanced logic/low power), **6** ($\approx$ 6,000 logic elements), **E22** (EQFP-144 package), **C8** (Commercial temperature grade 0°C to 85°C, speed grade 8), and **N** (Lead-free RoHS compliant).

3 Configuration & Programming Interfaces

The V3.0 board features two distinct 10-pin programming headers configured for use with Altera USB-Blaster download cables:

1. **JTAG (Joint Test Action Group) Interface**
2. **Active Serial (AS) Interface:** Writes configuration data directly into the onboard **EPCS16** (16 Mbit / 2 MByte) serial flash memory chip.

4 Onboard Peripherals & Pin Assignments

Peripheral Component	Board Label	FPGA Pin Assignment	Logic Standard / Notes
System Clock (50 MHz)	CLK50	PIN_23	3.3-V LVTTL (Global Clock Input)
User LEDs (Active Low)	LED1 LED2 LED3 LED4	PIN_87 PIN_86 PIN_85 PIN_84	3.3-V LVTTL (0 = Illuminated)
Push Buttons (Active Low)	KEY1 / S1 KEY2 / S2 KEY3 / S3 KEY4 / S4	PIN_88 PIN_89 PIN_90 PIN_91	3.3-V LVTTL (0 = Pressed, Debounce req.)
Piezo Buzzer	BUZZER	PIN_110	3.3-V LVTTL (1 = Tone Active)
7-Segment Anodes (Digit Sel.)	DIG1 (MSB) DIG2 DIG3 DIG4 (LSB)	PIN_133 PIN_132 PIN_131 PIN_128	3.3-V LVTTL (0 = Digit Enabled)
7-Segment Cathodes (Data)	SEG_A SEG_B SEG_C SEG_D SEG_E SEG_F SEG_G SEG_DP	PIN_127 PIN_126 PIN_125 PIN_124 PIN_121 PIN_120 PIN_119 PIN_115	3.3-V LVTTL (0 = Segment Lit)

Table 1: Quartus Prime physical pin mapping reference for Altera Cyclone IV V3.0 onboard peripherals.